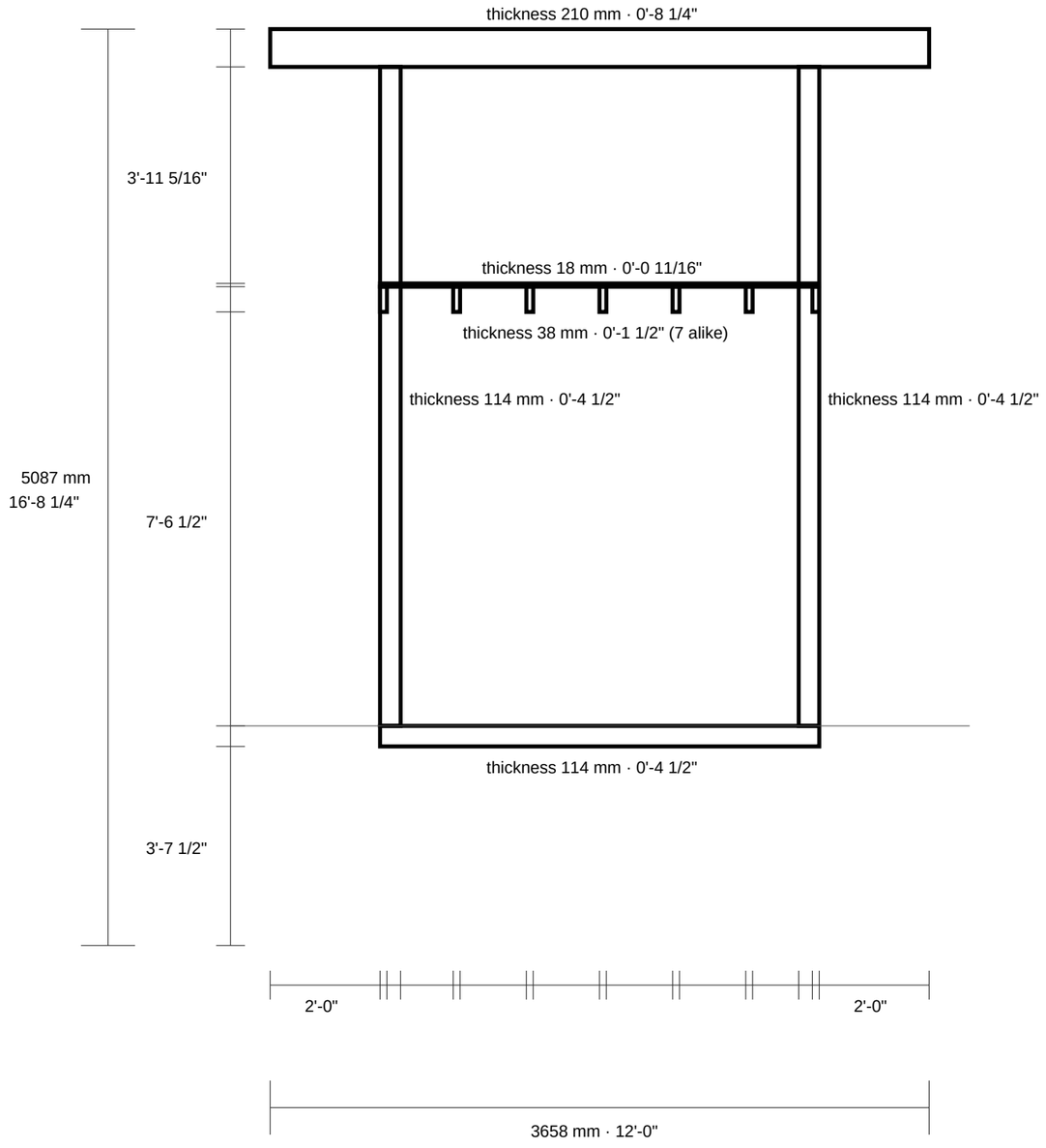

NOT MEASURED BY THE ENGINE

none

ASSUMED SETTINGS

floor.thickness_mm = 114 mm: assumed: a 4-1/2 in (114 mm) SIP floor panel, the wall standard's thickness, research row sip.prs610_panel_thickness; Jim's floor build-up gives it
weather.u_value = 2.8 w_per_m2_k: assumed: typical double glazing, not chosen by Jim
weather.solar_heat_gain_coefficient = 0.6 ratio: assumed: typical double glazing, not chosen by Jim
weather.inside_celsius = 20 celsius: assumed design day
weather.outside_celsius = 0 celsius: assumed design day
weather.sun_watt_per_square_metre = 500 w_per_m2: assumed clear-sky peak; the solver applies it to every side
finish.mean_reflectance = 0.5 ratio: assumed mid-grey interior; Jim's wall finish gives it
finish.sky_angle_degrees = 65 degree: assumed open sky; the site gives it
envelope.wall_r = 20 r_value: assumed R-20 for a SIP wall (ft2 F h per BTU); the SIP thickness gives it
envelope.roof_r = 30 r_value: assumed R-30; the roof build-up gives it
envelope.floor_r = 20 r_value: assumed R-20; the floor build-up gives it
envelope.door_r = 5 r_value: assumed R-5 for an insulated door, not checked against a source
heating.boiler_efficiency = 0.7 ratio: assumed 70 percent; sources run 55 to 80 percent
loft_floor.thickness_mm = 18 mm: assumed: 23/32 in (18 mm) combination subfloor, the IRC subfloor table's panel for a 24 in span, research row bom.subfloor_irc; at 16 in joist spacing a thinner panel may do
joists.member_depth_mm = 140 mm: assumed 2x6; the beam solver says whether it is enough
joists.spacing_mm = 406 mm: 16 in on centre, assumed
header.plies = 2 count: assumed: two plies side by side, 3-1/2 in of LVL or 3 in of 2x lumber, fit inside the SIP's 3-5/8 in core (sip.prs610_core_thickness)
roof_loads.roof_dead_load_psf = 15 psf: assumed: 15 psf, the roof dead load Boise's roof-only header look-up assumes, research row gl.hdr.vl_lookup_roof_only; the roof build-up gives it
roof_loads.deflection_limit_ratio = 240 ratio: assumed: span over 240, the live-load deflection limit of Boise's header tables, research row gl.hdr.vl_limits, applied to the whole load
roof_loads.roof_bears_on = compass_sides: assumed: the roof's 8 ft south hangover can only be carried by members running north-south across the south wall, so the roof bears on the south and north walls

PROJECT	SHEET	SHEET NUMBER	DATE
Small house on posts, from Jim's pencil sketch	**Section through the middle of the depth**	A-6 of 6	2026-10-20
CONFIG COMMIT	ENGINE PIN	SCALE	SOURCE
c2a9df0	we 02d17b7; rendered with we 073b001	1:32 (3/8" = 1'-0")	world-engine geometry; no line is drawn by hand